Mosairul Alam Tyhan

Aspiring Machine Learning Engineer

+8801406822246 tyhanhasan@gmail.com LinkedIn Website Mymensingh, Bangladesh

PROFESSIONAL SUMMARY

Aspiring Machine Learning Engineer with hands-on experience building ML applications using Python, Scikit-learn, TensorFlow, FastAPI, and Docker. Developed NewsCortex AI, an NLP-based news classification system using TF-IDF and an Artificial Neural Network (ANN), achieving 91% accuracy on 120K+ articles. Currently expanding my skills in Generative AI, LLM applications, and LangChain. Seeking ML/AI internships or entry-level roles to contribute to practical AI solutions and grow as an engineer.

EXPERIENCE

Machine Learning Intern 2026
Auspify Technologies *Remote*

- Developed a content-based recommendation system using TF-IDF and cosine similarity to recommend similar titles from a Netflix catalog of 8,000+ titles.
- Built and evaluated classification models, including Logistic Regression, Decision Trees, and Random Forest, etc for content type and audience rating prediction.
- Applied K-Means clustering with elbow and silhouette methods to identify three Netflix content segments based on genre patterns and content characteristics.
- Developed a time-series forecasting model using historical `date_added` data, time-based features, and lag variables to predict content additions over a 12-month horizon, achieving **R^2 = 0.89** through walk-forward validation.

Machine Learning Intern 2026 - Present
FlyRank *Remote*

- Complete weekly assignments and research tasks focused on AI concepts, machine learning fundamentals, and technical problem-solving.
- Participate in guided training and tutorials covering AI tools, effective AI usage, and introductory production-oriented workflows.

TECHNICAL SKILLS

Languages :	Python, SQL
ML/DL :	Scikit-learn, TensorFlow, Keras, Numpy, Pandas
Backend & Deployment :	FastAPI, Docker, Streamlit, Uvicorn, REST APIs, Streamlit Community Cloud
Tools & Database :	Git, GitHub, Jupyter, Kaggle, MySQL
ML Areas :	NLP, Deep Learning, Model Deployment

PROJECTS

Freight Rate Predictor | 2026

- Built an end-to-end freight rate prediction system, including preprocessing, feature engineering, and regression model comparison.
- Developed a FastAPI backend and Streamlit interface with input validation for real-time freight-rate predictions.
- Containerized and deployed the application with Docker, with a documented video walkthrough explaining the complete approach and implementation.

NewsCortex AI | 2026

- Built an NLP-based news classification system using TF-IDF, TensorFlow/Keras, and ANN (Deep Learning).
- Processed 120K+ articles across 4 categories with 10K TF-IDF features using unigram/bigram representations.
- Achieved 91% test accuracy and built a FastAPI backend, Dockerized the application, and deployed a Streamlit frontend for real-time predictions.

Heart Disease Risk Predictor | 2026

- Built a binary classification model using Logistic Regression to predict heart disease risk.
- Developed a FastAPI REST API with Pydantic validation for real-time predictions and probability scores.
- Dockerized and deployed the application with a Streamlit frontend, featuring feature-importance visualization and error handling.

EDUCATION

Diploma in Engineering in Computer Science & Technology
Mymensingh Polytechnic Institute
Expected Graduation: 2027

CERTIFICATION

Data Science & Machine Learning with Python
Ostad
Data Science Workshop for Research & Higher Studies
Ostad

LANGUAGES

- Bangla (Native)
- English (Working Proficiency)